- The Cloud Native Computing Foundation (CNCF) project's Fluentd (https://github.com/fluent/fluentd) and Fluent Bit (https://github.com/fluent/fluent-bit).
- The CNCF project OpenTelemetry offers an experimental log router called Collector (http://mng.bz/AoJK).
- Cloud provider–specific agents, such as Amazon CloudWatch agents (http://mng.bz/ZqJO), Azure Monitor agents (http://mng.bz/2DNd), or Google Cloud operations suite agents (https://cloud.google.com/monitoring/agent/).
- Proprietary agents, such as Elastic's Logstash (https://github.com/elastic/logstash) and Filebeat (http://mng.bz/1qEV), the Datadog agent (https://docs.datadoghq.com/agent/), or the Splunk Universal Forwarder (https://docs.splunk.com/Documentation/Forwarder).

To avoid exceedingly high costs in this phase, as well as in the destinations, you may wish to prune logs here. Some pruning strategies you might want to consider include

- *Sampling*—That is, dropping a certain percentage of the events, based on a heuristic
- *Filtering based on type*—For example, dropping `DEBUG` level logs in prod

There are commercial offerings, such as Cribl (https://cribl.com/), that can help define and apply pruning strategies at scale, across providers.

Destination costs of logs

Once logs from different microservices arrive at a central location (the destination, such as Splunk, CloudWatch, OpenSearch, or Datadog), the main cost type is storage. At scale, logs can be very expensive, so usually, you don't want to keep everything around forever. It is better to decide the retention period of logs based on the *data temperature*:

- *Hot data*—Your working set—that is, logs you're using in the current time frame to troubleshoot or develop. It has to be readily accessible and fast to query. Usually, that spans from days to weeks.
- *Warm data*—Something you might need but is not necessary immediately accessible or queryable. The retention period here might be months or even years.
- *Cold data*—All the (log) data you don't actively use. You may need to keep these logs lines around for regulatory reasons. It is best to store that kind of log data in cheap, reliable, and durable object storage, such as in Amazon S3 Glacier. It may take minutes or hours to access the data, but given that you seldom, if at all, use it, that's a tradeoff you can live with.

Benefits of logs

Now that you have an idea about the costs of logs, let's briefly talk about their benefits. First off, logs are universally adopted and understood. Every developer knows how to emit log lines, and there's fantastic support for logs in every programming language,